No.

2-oy: 9-dars: Model Improvement

y_pred y_test (y_actual)

- Model Prediction qiymatini xaqiqiy qiymatga yaqinlashtirish

Qachon ishimiz tugadi: qachonki maksimal error/darajada juda katta bolib ketsa.

- Regression
  - ↳ Reducing Error (mse, mae)
  - ↳ Increasing r2-score

- Classification
  - ↳ Increasing accuracy score

Model Improvement yollari:

1. Data quality and quantity
   - Koproq dataset collection
   - Datasetni tozalang (Noisy tozalash kerak: yod element, bosh space)
2. Feature Engineering
   - Yangi featurelar yaratish
   - Transform features
   - Domain knowledge
   - Feature selection

3. Scaling (MinMaxScaler, StandardScaler, RobustScaler)
4. Turli xil algorithmlar b/n tekshirish
   - Bazi xollarda sodaroq algoritmlar xam yuqori natija beradi (Bu datani Input va Output qiymatlar orasidagi munosabatlarga boglig)

5. Hyperparametr tuning
- Bizning algoritm va dataset ga mos parametr/ni topish jarayoni

6. Cross validation
7. Regularization
8. Ensemble methods → kuchliroq algoritmlar
9. Handling Imbalanced Dataset (Classification)
[900 - 0 va 100 - 1 bolsa o chiqishiga moyillig oshadi shuni dal qilish (overfitting, underfitting)]
→ Balansga keltirish bo'ladi

Yaxshi Model Formulasi

Smart features

Yaxshi model = Yaxshi dataset + Aqlli va foydali features + to'g'ri algorithmlar + tunning + overfittingni oldini olish.

Transform Features

- Input feature larni ML algoritmlar u-n moslashtirish jarayoni.
- Yani featurelarni kompyuterga osonroq tilda tushuntirish.
- Qisqacha: Featurelarni o'zgartirish.

CamScanner

Transform features: Turlari

- Sqrt
- Feature scaling
- Log Transformation

} 3ta — Qiymatlar juda katta bolsa qollanil

- Power Transformation
  ↳ x^2, x^3 $x^2 = Age^2$, $x^3 = Age^3$

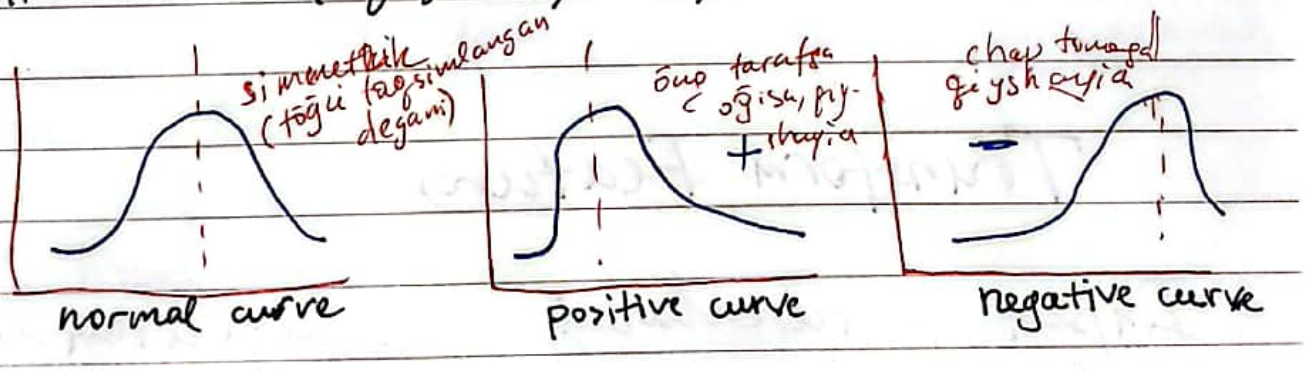
(mantiqsiz bolishi mumkin lekin sun model rivojlantirish usullaridan bizi xisoblanadi)

→ Qiymatlar juda kichik bolsa.

- Polynomial Features (bir haddan koprhadga otish jarayoniga aytiladi)
- Encoding (Objectni - floatga, objectni - intga)

Transform features: Log Transformation

- skewness (qiyshayish)



- Matematik transformation
  ↳ Bunda faqatgina biz Log transformationni ishlata olamiz qachongi bizda positive skewness bolsa.
  ↳ Negative skewness bolsa ($x^2, x^4, \dots x^n$ sqr transformation)

No.

Danak darajali transformation usuli bu positive tomonga yani musbat tomonga qiyshayib ketgan qiymatlarni, raw datalarni to'g'rilash u-n, yani ilaji bo'ycha normal holatga keltirish u-n ishlatilinadigan usul hisoblanadi

- Faqatgina musbat qiymatlarda ishlaydi (manfiy bolsa barcha qiymatlarga c = constant qiymat qo'shing)
  ↳ bu agar manfiy bolsa 1. { skewness manfiy bolsa

Power transformation Polynomial Transformation

qo'yamiz

2. Yoki bitta constant qiymat qo'shishimiz kerak.

- Asosan Positive skewnesslik xolatlarda ishlaydi va maqsad qiymatlarni normallashtirish.

- Tree - based algorithmlar bazi xolatlarda foydasiz va keraksiz.

▲ num_col = df.select_dtypes(include=['float64'])

1. skew <0.5 --- Transformation kerak emas
num_col larni qiyshayish darajalarini aniqlash
▲ num_col.skew()

#2 skew <0. --- squared transformation

#3 skew >= 0.5 --- log transformatsiyani qo'llashimiz k-k

▲ import numpy as np
filter qilish orqali log ga tushadiganlarini aniqladik

▲ log_transformation = skewness[(skewness >=0.5)].index.tolist()

↳ ['id', 'goals', ...]

bizga hamma 0,5 teng yoki katta larini chiqarib beradi.. Yani log transformatsiyaga qo'yadigan ustunlarimizni aniqlab oldik.

log ni qo'llaymiz

```
for col in log_transformation:
    df[col + '_log'] = np.log1p(df[col])
```

```
df.info()
```

↳ boshida bizda 18 ta edi.. Endi esa bizda 31 ta bo'lib qoldi. id_log, ..., x6Chain_log, x6Building_log